

A Bounded Collapsed Almost Uncollapsed Map

Automated discovery packet

June 14, 2026

Source question

Braga's paper [1] asks Problem 2.6: "Is there a map $X \rightarrow Y$ which is collapsed, almost uncollapsed and bounded (in particular not solvent), for some Banach spaces X and Y ?"

Hence, $\|b(2^{2^n})\| \rightarrow 0$, as $k \rightarrow \infty$. So, b is collapsed. $\square$

Problem 2.6. Is there a map $X \rightarrow Y$ which is collapsed, almost uncollapsed and bounded (in particular not solvent), for some Banach spaces X and Y ?

Definitions used

For a map $f : (M, d) \rightarrow (N, \partial)$, the source defines

$$\rho_f(t) = \inf\{\partial(f(x), f(y)) : d(x, y) \geq t\}$$

and

$$\bar{\rho}_f(t) = \inf\{\partial(f(x), f(y)) : d(x, y) = t\}.$$

The map is uncollapsed if $\rho_f(t) > 0$ for some $t > 0$, and almost uncollapsed if $\bar{\rho}_f(t) > 0$ for some $t > 0$. In the terminology of the paper, collapsed means not uncollapsed.

Proof Intuition

The exact-distance requirement in almost uncollapsedness is much weaker than the large-distance requirement in uncollapsedness. A periodic map into the unit circle separates all pairs at one chosen distance, for instance distance π , but repeats values at arbitrarily large distances. This gives almost uncollapsedness and collapsedness at the same time. Since the circle is bounded, the same map is automatically not solvent. If one wants the domain to be infinite-dimensional, the same idea can be made set-theoretic: identify points modulo a large-period cyclic subgroup and send different cosets to different unit vectors.

Result

Theorem 1. *There are Banach spaces X, Y and a bounded map $f : X \rightarrow Y$ which is collapsed and almost uncollapsed. In particular, Problem 2.6 of [1] has an affirmative answer. Moreover, if the problem is read as asking for an infinite-dimensional domain, X can be chosen infinite-dimensional.*

Proof. Take $X = \mathbb{R}$, take $Y = \mathbb{R}^2$ with its Euclidean norm, and define

$$f(x) = (\cos x, \sin x).$$

The map is bounded, since $\|f(x)\|_2 = 1$ for every $x \in \mathbb{R}$.

It is almost uncollapsed. Indeed, if $|x - y| = \pi$, then $y = x + \pi$ or $y = x - \pi$, and hence $f(y) = -f(x)$. Therefore

$$\|f(x) - f(y)\|_2 = 2$$

for every pair $x, y \in \mathbb{R}$ with $|x - y| = \pi$. Thus

$$\bar{\rho}_f(\pi) = 2 > 0.$$

It is collapsed. Let $t > 0$. Choose an integer $k \geq 1$ such that $2\pi k \geq t$. Then $|2\pi k - 0| \geq t$, while

$$f(2\pi k) = f(0).$$

Consequently $\rho_f(t) = 0$ for every $t > 0$. Thus f is not uncollapsed, i.e. it is collapsed in the source terminology.

Finally, f is not solvent. Its image has diameter at most 2, so for any $n > 2$ no interval of domain distances can force all image distances to be greater than n . This proves the claimed bounded collapsed almost uncollapsed example.

For the infinite-dimensional variant, let X be any infinite-dimensional Banach space and choose $u \in X$ with $\|u\| = 1$. Let

$$A = 2\pi\mathbb{Z}u$$

and let $I = X/A$ be the set of cosets of the additive subgroup A . Put $Y = \ell_2(I)$, with its standard unit vectors $(e_i)_{i \in I}$, and define

$$F(x) = e_{x+A}.$$

This map is bounded, since every value has norm 1. If $\|x - y\| = \pi$ and $x + A = y + A$, then $x - y = 2\pi ku$ for some $k \in \mathbb{Z}$, so $\|x - y\| = 2\pi|k|$, impossible. Hence exact-distance- π pairs always belong to different cosets, and therefore

$$\|F(x) - F(y)\|_2 = \sqrt{2} \quad (\|x - y\| = \pi).$$

Thus $\bar{\rho}_F(\pi) = \sqrt{2} > 0$. On the other hand, for every $t > 0$ one can choose $k \geq 1$ with $2\pi k \geq t$, and then

$$\|2\pi ku - 0\| \geq t, \quad F(2\pi ku) = F(0).$$

So $\rho_F(t) = 0$ for every $t > 0$, and F is collapsed. Its image has diameter $\sqrt{2}$, so it is not solvent. This variant is not claimed to be continuous; it is included only to remove any possible finite-dimensional reading concern. $\square$

Verification notes

The first proof uses only the source definitions and the elementary identities $f(x + \pi) = -f(x)$ and $f(x + 2\pi k) = f(x)$. The infinite-dimensional variant uses only cosets of the cyclic subgroup $2\pi\mathbb{Z}u$. The script `code/check_circle_map.py` performs a finite numerical smoke check of these identities and the boundedness normalization; it is not needed for the proof. The script `code/make_open_problem_crop.py` recreates the source screenshot from `source_paper.pdf`.

Verifier focus: confirm that the paper uses “collapsed” as the negation of “uncollapsed”, as indicated in the surrounding text before Proposition 2.5, and that Problem 2.6 does not impose any additional regularity such as being a cocycle or a coarse map. The finite-dimensional displayed example is in fact Lipschitz, but it is not coarse-embedding-like because it is bounded. The infinite-dimensional variant answers only the unrestricted-map reading.

Novelty and literature check

Local duplicate check on June 14, 2026 searched `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` for 1607.06127, title keywords, `collapsed`, `almost uncollapsed`, `bounded`, and related terms; no matching packet or attempt was found.

Web searches on June 14, 2026 used exact and close phrase searches for:

- collapsed, almost uncollapsed, bounded map, and Banach spaces;
- the source title together with almost uncollapsed and collapsed;
- the exact Problem 2.6 wording and shorter variants.

They found the source paper but no separate later paper explicitly answering this problem. Novelty confidence is moderate rather than high because the example is elementary and may be folklore.

Human review recommendation

Send to human as a likely valid full answer to Problem 2.6. The main semantic point to check is whether the author intended an unstated restriction to cocycles or to another class of maps; under the printed statement, the circle map answers the question.

References

- [1] B. M. Braga, *On weaker notions of nonlinear embeddings between Banach spaces*, arXiv:1607.06127, 2016.

Human Review: Bounded Collapsed Almost Uncollapsed Map

Original automatic proof: `solution_packet.pdf`

Checked date: 15 June 2026

Status: Verified.

Verdict: The proof is correct. Under the printed wording of Problem 2.6, the circle map gives a valid solution.

Human Comments

The source problem asks whether there is a map $X \rightarrow Y$, for Banach spaces X and Y , which is collapsed, almost uncollapsed, and bounded. The proposed example

$$f : \mathbb{R} \rightarrow \mathbb{R}^2, \quad f(x) = (\cos x, \sin x)$$

does exactly this.

Boundedness is immediate, since the image lies in the Euclidean unit circle. The map is collapsed because it is periodic: for every $t > 0$, choose $k \in \mathbb{N}$ with $2\pi k \geq t$. Then

$$|2\pi k - 0| \geq t, \quad f(2\pi k) = f(0),$$

so the compression modulus satisfies

$$\rho_f(t) = 0$$

for every $t > 0$. Hence f is not uncollapsed, i.e. it is collapsed in the terminology of the paper.

On the other hand, f is almost uncollapsed. If $|x - y| = \pi$, then $f(y) = -f(x)$, and therefore

$$\|f(x) - f(y)\|_2 = 2.$$

Thus the exact compression modulus satisfies

$$\bar{\rho}_f(\pi) = 2 > 0.$$

So f is almost uncollapsed.

The construction also makes clear why the example works: periodicity produces arbitrarily far apart points with identical images, while the antipodal distance π gives uniform separation at one exact distance. In this sense, the example is not accidental; any suitably chosen periodic map with a uniformly separated exact distance would give the same phenomenon.

Infinite-Dimensional Variant

The infinite-dimensional variant in the packet is also conceptually sound. One chooses a unit vector $u \in X$, quotients by the cyclic subgroup $2\pi\mathbb{Z}u$, and maps cosets to distinct unit vectors in an ℓ_2 -space. This recreates the same periodic-collapse mechanism along one direction while preserving separation at exact distance π . This variant is not needed for the printed problem, since $X = \mathbb{R}$ is already a Banach space, but it addresses any possible preference for infinite-dimensional domains.

Cocycle Comment

Reading around Problem 2.6, I do not see any explicit requirement that the map be a cocycle or arise from an affine isometric action. The problem is stated for a “map” $X \rightarrow Y$, and the finite-dimensional circle example is even continuous and Lipschitz. Therefore it answers the printed question.

In fact, the finite-dimensional example can also be written as a continuous bounded cocycle. Let R_t denote rotation by angle t on $\mathbb{R}^2$, and define

$$b(t) = R_t e_1 - e_1 = (\cos t - 1, \sin t).$$

Then b is a cocycle for the isometric representation $t \mapsto R_t$, since

$$b(s+t) = b(s) + R_s b(t).$$

Subtracting the constant vector e_1 does not change pairwise distances, so this cocycle has the same collapsed, almost uncollapsed, and bounded behaviour as the circle map.

The infinite-dimensional coset variant can also be written as an algebraic bounded cocycle. If $A = 2\pi\mathbb{Z}u$, $I = X/A$, and $Y = \ell_2(I)$, let X act on Y by permuting the basis vectors according to coset translation:

$$\pi_x e_{[y]} = e_{[x+y]}.$$

Then

$$b(x) = \pi_x e_{[0]} - e_{[0]} = e_{[x]} - e_{[0]}$$

is a bounded cocycle. It has the same collapse and exact-distance separation properties as the coset map. However, this infinite-dimensional cocycle is not continuous in general; arbitrarily small nonzero $x \notin A$ already produce a jump in the image.

Review Outcome

Accepted as a valid solution to Problem 2.6 as printed.